

Smart Metro Ticket Booking System

IDEATION PHASE

Date	
Team ID	
Project Name	Smart Metro Ticket Booking System
Maximum Marks	

3. Brainstorming and Idea Prioritization

3.1 Objective

A team brainstorming session was conducted to generate possible solution ideas for the problem of manual, queue-dependent metro ticket booking. Ideas were then grouped, evaluated, and prioritized based on feasibility on the ServiceNow platform, customer impact, and implementation effort.

3.2 Raw Brainstormed Ideas

- Digital ticket booking portal built on ServiceNow Service Catalog with station and travel-date selection.
- Automated fare calculation engine based on source/destination distance and passenger count using Flow Designer.
- QR-code based digital ticket generation for entry/exit scanning at gates.
- SMS/Email/Push notifications for booking confirmation and ticket delivery.
- Wallet/UPI/card payment gateway integration for instant payment.
- Season pass / monthly pass booking and auto-renewal reminders.
- Live queue and crowd-status display integrated with the booking catalog item.
- Self-service ticket cancellation and refund workflow.
- Chatbot/Virtual Agent for booking assistance and FAQs.
- Admin dashboard for metro staff to monitor bookings, revenue, and station-wise traffic.
- Accessibility mode with voice assistance for elderly/differently-abled commuters.
- Group booking option for families or corporate travel.
- Integration with station gate hardware/IoT scanners for automatic QR validation.
- Analytics and reporting module for peak-hour prediction and fare optimization.

3.3 Idea Grouping (Affinity Mapping)

Theme	Grouped Ideas
Core Booking Experience	Service Catalog booking form, station/date/passenger selection

Theme	Grouped Ideas
Communication & Delivery	Notifications (SMS/Email/Push), booking confirmation
Payments	UPI/Card/Wallet integration, season pass renewal
Operations & Admin	Admin dashboard, analytics/reporting, gate hardware integration
Support & Accessibility	Virtual Agent/chatbot, voice-assisted accessibility mode, cancellation

3.4 Prioritization Matrix (Impact vs. Effort)

Idea	Customer Impact	Implementation Effort	Priority
Service Catalog based ticket booking form	High	Low	P0 - Must Have
Automated fare calculation (Flow Designer)	High	Medium	P0 - Must Have
QR-code digital ticket generation	High	Low	P0 - Must Have
Booking confirmation notifications (SMS/Email)	High	Low	P0 - Must Have
Payment gateway integration	High	Medium	P1 - Should Have
Ticket cancellation/refund workflow	Medium	Medium	P1 - Should Have
Admin dashboard for staff/operations	Medium	Medium	P1 - Should Have
Season pass booking & renewal	Medium	Medium	P2 - Could Have
Virtual Agent/chatbot support	Medium	High	P2 - Could Have
Gate hardware/IoT scanner integration	Medium	High	P2 - Could Have
Accessibility voice-assistance mode	Medium	High	P2 - Could Have
Analytics & peak-hour prediction	Low	High	P3 - Future Scope

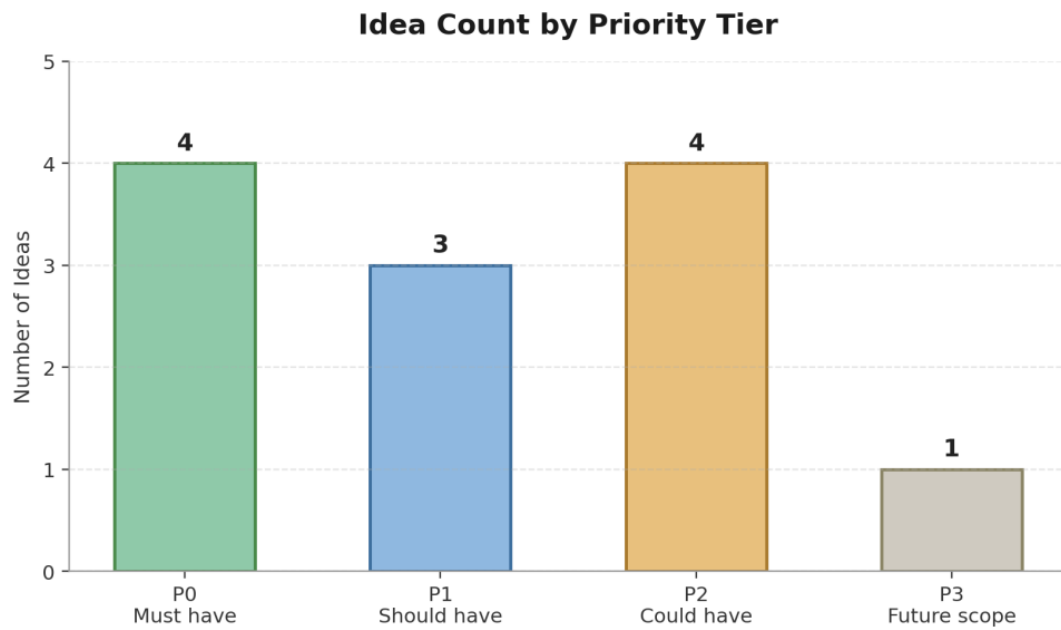


Figure 1: Number of brainstormed ideas per priority tier

3.5 Finalized Ideas for Phase 1 (MVP)

Based on the prioritization matrix, the following ideas were selected as the Minimum Viable Product (MVP) scope for this project, all of which are achievable using core ServiceNow components:

- Service Catalog item for selecting source station, destination station, travel date, and number of passengers.
- Flow Designer workflow to automatically calculate the fare based on station distance/zone and passenger count.
- Automatic generation of a unique digital ticket with an embedded QR code upon successful booking.
- Notification module to instantly send the digital ticket and booking confirmation to the commuter.
- Basic booking history/status tracking for each user through the Service Catalog request record.